

1.		4
1.1.		4
1.2.		4
2.		5
2.1.		5
2.2.		5
2.3.		5
2.4.		6
2.5.		6
2.6.		6
2.7.		6
2.8.		7
3.		8
3.1.		8
3.2.		8
3.3.		8
3.4.		9
3.5.		9
3.6.		11
3.7.		11
3.8.		11
3.9.		12
3.10.		13
3.11.		13
3.12.		13
3.13.		14

3.14.		14
3.15.		14
3.16.		15
3.17.		15
3.18.		16
3.19.		17
3.20.		17
3.21.		18
3.22.		19
3.23.		19
4.		20
4.1.		20
4.2.		20
4.3.		21
4.4.		22
4.5.		23
4.6.		24
4.7.		24
4.8.		25
4.9.		26
4.10.		27
4.11.		27
5.		29
5.1.		29
5.2.		29
5.3.		29
5.4.		30
5.5.		31
5.6.		31
5.7.		32
5.8.		33

6.		35
6.1.		35
6.2.		35
6.3.		36
6.4.		36
6.5.		37
6.6.		37
6.7.		38
6.8.		39
6.9.		40
6.10.		40
6.11.		42
6.12.		43
6.13.		45
6.14.		46
7.		48
7.1.		48
7.2.		48
7.3.		49
7.4.		49
7.5.		49
7.6.		49
7.7.		50
7.8.		50
7.9.		50
7.10.		50
7.11.		51
7.12.		51
7.13.	2	51
7.14.		52
7.15.		52

1.

1.1.

2022

pp. 40 - 41

1.2.

2022

pp. 43, 47 - 54

12 cm² g

(1) = +

(2)

(3)

(,) = (11, 23), (23, 49), (5, 10), (50, 115), (32, 64), (21, 45)
(16, 35), (39, 84), (25, 50), (45, 100), (5, 11), (34, 70)

2.1.

(1)

2.2.

- (1)
- (2)
- (3)
- (4) 3

1

$$(\quad = \quad) = \quad \times \quad (\quad = 1, 2, 3, 4, 5, 6) \quad \times$$

- $$\begin{aligned} (1) \quad & \\ (2) \quad & = 1, 2, 3, 4, 5, 6 \quad (=) \\ (3) \quad & [] \\ (4) \quad & (\geq 4) \end{aligned}$$

2.4.

2023

pp. 81, 86

$$f(x) = \begin{cases} \frac{x}{3} & (x \geq 10) \\ 0 & (x < 10) \end{cases}$$

(1)

(2) $f(f(x))$

(3) $f(x) \geq \frac{1}{2}$

2.5.

2024

pp. 81, 86

$$f(x) = \begin{cases} (1-x) & (0 \leq x \leq 1) \\ 0 & (\text{otherwise}) \end{cases}$$

(1)

(2) $\left(\frac{1}{2} < x \leq \frac{2}{3}\right)$

(3) $f(f(x))$

(4) $\left(\frac{1}{2} \leq x < 3\right)$

2.6.

2022

2023

2024

pp. 134, 151, 180

(1)

(2)

(3)

(4)

2.7.

2023

2023

pp. 134, 151, 180

$1, 2, \dots,$

2

(1) $1, 2, \dots,$

(2) $2 /$

(3)

(4)

(5)

(6)

1

2.8.

2025

(1)

2

(2) 200

60

6

42

78

3.

3.1.

2024

p. 107

6

4

1

2

(1) 2

(2) 2

$$(1) \frac{6}{10} \frac{5}{9} = \frac{1}{3}$$

$$(2) \frac{6}{10} \frac{5}{9} + \frac{4}{10} \frac{6}{9} = \frac{3}{5}$$

3.2.

2026

p. 107

3

5

(1)

1

1

(a) 2

(b) 2

(c) 2

1

(2)

2

1

(3)

(1)

(2)

$$(1) (a) \frac{3}{8} \frac{2}{7} = \frac{6}{56} = \frac{3}{28}$$

$$(b) \frac{3}{8} \frac{2}{7} + \frac{5}{8} \frac{3}{7} = \frac{21}{56} = \frac{3}{8}$$

$$(c) \frac{6/56}{21/56} = \frac{6}{21} = \frac{2}{7}$$

$$(2) \frac{3}{8}$$

$$(3) 1$$

3.3.

2022

pp. 107 - 108

$$\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 1 & 2 & & \end{array} = \begin{vmatrix} 1 & - & 2 \end{vmatrix}$$

(1)

(2)

(3)

$$(1) \quad \begin{array}{c|cccc|c} & 0 & 1 & 2 & 3 & \\ \hline & 1/4 & 3/8 & 1/4 & 1/8 & 1 \end{array}$$

$$(2) \quad [] = \frac{1}{4} \cdot 0 + \frac{3}{8} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 = \frac{5}{4}$$

$$(3) \quad \text{Var}[] = []^2 - ([])^2 = \left(\frac{1}{4} \cdot 0^2 + \frac{3}{8} \cdot 1^2 + \frac{1}{4} \cdot 2^2 + \frac{1}{8} \cdot 3^2 \right) - \left(\frac{5}{4} \right)^2 = \frac{5}{2} - \frac{25}{16} = \frac{15}{16}$$

3.4.

2023

2025

pp. 113 - 114

(,)

$$(,) = \begin{cases} 6^{-2-3} & (\geq 0, \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

$$\geq 0 \quad \geq 0$$

$$_1() = \int_{-\infty}^{\infty} (,) = \int_0^{\infty} 6^{-2-3} = 6^{-2} \left[-\frac{1}{3}^{-3} \right]_0^{\infty} = 2^{-2}$$

$$_2() = \int_{-\infty}^{\infty} (,) = \int_0^{\infty} 6^{-2-3} = 6^{-3} \left[-\frac{1}{2}^{-2} \right]_0^{\infty} = 3^{-3}$$

$$_1() = \begin{cases} 2^{-2} & (\geq 0) \\ 0 & (< 0) \end{cases} \quad _2() = \begin{cases} 3^{-3} & (\geq 0) \\ 0 & (< 0) \end{cases}$$

$$\geq 0 \quad \geq 0$$

$$(,) = _1() _2()$$

3.5.

2026

$$\int_0^{\infty} e^{-x} dx = 1, \quad \int_0^{\infty} e^{-y} dy = 1$$

$$(1) \quad f(x, y) = \begin{cases} e^{-x-y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

(a) $f(x, y) = \dots$ (b) $f(x, y) = \dots$

(b)

$$(2) \quad f(x, y) = \begin{cases} (x+y)e^{-x-y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

(a) $f(x, y) = \dots$ (b) $f(x, y) = \dots$

(1) (a) $x \geq 0, y \geq 0$

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^{\infty} e^{-x-y} dy = e^{-x} \int_0^{\infty} e^{-y} dy = e^{-x}$$

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^{\infty} e^{-x-y} dx = e^{-y} \int_0^{\infty} e^{-x} dx = e^{-y}$$

$$f(x) = \begin{cases} e^{-x} & (x \geq 0) \\ 0 & (\text{otherwise}) \end{cases}, \quad f(y) = \begin{cases} e^{-y} & (y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

(b) $f(x, y) = e^{-x-y} = f(x) f(y)$

(2) (a) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

$$\int_0^{\infty} \int_0^{\infty} (x+y)e^{-x-y} dx dy = \int_0^{\infty} e^{-y} dy \int_0^{\infty} x e^{-x} dx + \int_0^{\infty} e^{-x} dx \int_0^{\infty} y e^{-y} dy = 2$$

$$= \frac{1}{2}$$

(b) $x \geq 0, y \geq 0$

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy = \frac{1}{2} e^{-x} \int_0^{\infty} e^{-y} dy + \frac{1}{2} e^{-x} \int_0^{\infty} y e^{-y} dy = \frac{1}{2} (1 + 1) e^{-x}$$

$$\begin{aligned} (\quad) &= \int_{-\infty}^{\infty} (\quad , \quad) = \frac{1}{2} - \int_0^{\infty} \underset{=1}{\text{ssst}} + \frac{1}{2} - \int_0^{\infty} \underset{=1}{\text{ssst}} = \frac{1}{2} (\quad + 1) - \\ &= \quad = 0 \quad (\quad , \quad) \neq (\quad) (\quad) \end{aligned}$$

3.6.

2022

pp. 103 - 104, 113

$$\begin{aligned} 1 \qquad \qquad \qquad 4 \qquad \qquad \qquad 2 \qquad \qquad \qquad 2 \qquad \qquad \qquad 1 \\ &= 0 \qquad \qquad \qquad = 1 \qquad \qquad \qquad 2 \\ &= 0 \qquad \qquad \qquad = 1 \end{aligned}$$

(1) (\quad , \quad)

(2)

(1)

	0	1	
0	$\frac{2}{5}$	$\frac{2}{15}$	$\frac{2}{3}$
1	$\frac{4}{15}$	$\frac{1}{15}$	$\frac{1}{3}$
	$\frac{2}{3}$	$\frac{1}{3}$	1

(2) $(\quad = 0, \quad = 0) \neq (\quad = 0) (\quad = 0)$

3.7.

2022

pp. 67, 114 - 115

2

$$\begin{aligned} (\quad \cap \quad) &= (\quad) (\quad) \\ (\quad \cap \quad) &= 0 \qquad \qquad \qquad 2 \\ &2 \end{aligned}$$

3.8.

2022

2023

2026

p. 115

3

$\frac{1}{2}$

(1)

1

2

(2)

1

2

2

3

(3)

(|)

(1)

$$\Omega = \begin{pmatrix} & & \end{pmatrix}^3 \qquad \qquad \qquad \begin{pmatrix} & & \end{pmatrix}$$
$$\begin{matrix} 1 & 2 & 3 \end{matrix} \qquad \qquad \qquad \equiv \qquad \qquad \qquad \Omega$$
$$\qquad \qquad \qquad \equiv \qquad \qquad \qquad \Omega_0 = [\begin{pmatrix} & & \end{pmatrix}]_{\equiv}$$
$$\Omega_1 = [\begin{pmatrix} & & \end{pmatrix}]_{\equiv} \quad \Omega_2 = [\begin{pmatrix} & & \end{pmatrix}]_{\equiv} \quad \Omega_3 = [\begin{pmatrix} & & \end{pmatrix}]_{\equiv} \qquad \qquad \qquad |\Omega_0| = \begin{pmatrix} 3 \\ 0 \end{pmatrix} =$$
$$1 \quad |\Omega_1| = \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 3 \quad |\Omega_2| = \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 3 \quad |\Omega_3| = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 1 \qquad \qquad \qquad |\Omega| = 8$$
$$\qquad \qquad \qquad \frac{4}{7/8} = \frac{4}{7}$$

(2)

$$= \{ \begin{pmatrix} & & \end{pmatrix}, \begin{pmatrix} & & \end{pmatrix} \} \qquad \qquad \qquad = \{ \begin{pmatrix} & & \end{pmatrix}, \begin{pmatrix} & & \end{pmatrix} \}$$
$$\cap \qquad \cap \qquad \cap = \{ \begin{pmatrix} & & \end{pmatrix} \} \qquad \qquad \qquad |\Omega| \qquad \qquad \qquad \begin{pmatrix} \end{pmatrix} = \frac{1}{4} \quad \begin{pmatrix} \end{pmatrix} =$$
$$\frac{1}{4} \quad \begin{pmatrix} \cap \end{pmatrix} = \frac{1}{8} \qquad \qquad \qquad \begin{pmatrix} \cap \end{pmatrix} =$$
$$\begin{pmatrix} \end{pmatrix} \begin{pmatrix} \end{pmatrix}$$

(3)

$$\begin{pmatrix} | \end{pmatrix} = \frac{\begin{pmatrix} \cap \end{pmatrix}}{\begin{pmatrix} \end{pmatrix}} = \frac{1/8}{1/4} = \frac{1}{2}$$

3.9.

2

4

2

7

2

9

(1)

(2)

2

= 4

+

= 7

+

= 9

(1)

•

() = $\frac{1}{6}$

$$\begin{aligned}
 & \bullet \quad () = (+ = 7) = \frac{1}{6} \\
 & \bullet \quad (\cap) = (= 4, + = 7) = (= 4, = 3) = \frac{1}{36} \\
 & \quad (\cap) = () () \\
 (2) \quad & \bullet \quad () = \frac{1}{6} \\
 & \bullet \quad () = (+ = 9) = \frac{1}{9} \\
 & \bullet \quad (\cap) = (= 4, + = 9) = (= 4, = 5) = \frac{1}{36} \\
 & \quad (\cap) \neq () ()
 \end{aligned}$$

3.10.

2023

pp. 72 - 73, 109 - 110, 114 - 115

2

$$() = () =$$

$$(1) \quad ()$$

$$(2) \quad (\cap \mid)$$

$$(1) \quad () = + -$$

$$(2) \quad (\cap \mid) = \frac{(\cap)}{()} = \frac{-}{+ -}$$

3.11.

2025

1 5

$$= -$$

$$(1)$$

$$(2)$$

$$(3)$$

$$(4)$$

3.12.

2025

pp. 118 - 119

$$(1) \quad 2$$

(2)

500

1

2

1

2

3 : 1

0.98 0.90

0.01

1

1, 2

3.13.

2022

2023

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3

5

1

2

(1)

1

(2)

2

1

3.14.

2023

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1

10%

3

(1)

(2)

2

(1)

$\frac{9}{10} \frac{9}{10} \frac{9}{10} = 0.7290$

(2)

$\frac{\frac{9}{10} \frac{1}{10}}{\frac{1}{10} + \frac{9}{10} \frac{1}{10} + \frac{9}{10} \frac{1}{10} \frac{1}{10}} = \frac{90}{271} \approx 0.3321$

3.15.

2022

pp. 118 - 119

5

4

3

1

100 g

45 g

30 g

(1)

1

40 g

$$(2) \quad \frac{2}{100 \text{ g}} = \frac{2}{2}$$

$$(3) \quad \frac{3}{150 \text{ g}} = \frac{3}{3}$$

$$(1) \quad \frac{\binom{5}{1}}{\binom{12}{1} - \binom{3}{1}} = \frac{5}{9}$$

$$(2) \quad \frac{\binom{5}{2}}{\binom{12}{2} - \binom{7}{2}} = \frac{10}{66 - 21} = \frac{2}{9}$$

$$(3) \quad \frac{\binom{5}{3}}{\binom{12}{3} - \binom{7}{3}} = \frac{10}{220 - 35} = \frac{10}{185} = \frac{2}{37}$$

3.16.

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$$\frac{A}{A} = \frac{0.1\%}{99\%} = \frac{10\%}{10\%}$$

$$(1) \quad \frac{A}{A} = \frac{1}{1}$$

$$(2) \quad \frac{A}{A} = \frac{1}{1}$$

$$(1) \quad \frac{\binom{A}{1}}{\binom{A}{1} + \binom{2}{1}} = \frac{0.001 \cdot 0.99}{0.001 \cdot 0.99 + 0.999 \cdot 0.10} = \frac{99}{10089} \approx 0.0098$$

$$(2) \quad \frac{\binom{A}{2}}{\binom{A}{2} + \binom{2}{2}} = \frac{0.001 \cdot 0.99^2}{0.001 \cdot 0.99^2 + 0.999 \cdot 0.10^2} = \frac{9801}{109701} \approx 0.0893$$

3.17.

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$$\frac{99\%}{99\%} = \frac{0.5\%}{0.5\%}$$

$$1$$

(1)

(2)

(3)

(4)

$$(1) \quad (n) = () (|) = 0.005 \cdot 0.99 = 0.00495$$

$$(2) \quad (n) = () (|) = 0.995 \cdot 0.01 = 0.00995$$

$$(3) \quad () = (n) + (n) = 0.00495 + 0.00995 = 0.0149$$

$$(|) = \frac{(n)}{()} = \frac{0.00995}{0.0149} \approx 0.6678$$

$$(4) \quad (|) = \frac{(n)}{()} = \frac{() (|)}{1 - ()} = \frac{0.005 \cdot 0.01}{0.9851} \approx 0.0001$$

3.18.

2023

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	S			S
	K			
•	S	K	99%	1%
•	S	K	99%	1%

$$(1) \quad K \quad S$$

$$(2) \quad (1)$$

(1)

$$(|) = \frac{() (|)}{() (|) + () (|)} = \frac{\frac{1}{10000} \cdot \frac{99}{100}}{\frac{1}{10000} \cdot \frac{99}{100} + \frac{9999}{10000} \cdot \frac{1}{100}} = \frac{99}{10098} \approx 0.0098$$

(2)

3.19.

2025

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A

B

$$(1) \quad (A|B) = 0.98 \quad (B|A) = 0.005 \quad (A) = 0.001$$

$$(2) \quad (A|B) = 0.003 \quad (B) = 0.0005 \quad (A|B) \geq 0.142 \quad (A|B)$$

(1)

$$(A|B) = \frac{(A) (B|A)}{(A) (B|A) + (B) (A|B)} = \frac{0.001 \cdot 0.98}{0.001 \cdot 0.98 + 0.999 \cdot 0.005} = \frac{980}{5975} \approx 0.1640$$

(2) (1)

$$(A|B) = \frac{0.0005 (B|A)}{0.0005 (B|A) + 0.9995 (A|B)} \geq 0.142$$

$$(A|B) \geq \frac{1}{0.0005} \frac{1}{1 - 0.142} + 0.142 \cdot 0.9995 \cdot 0.003 \approx 0.9925$$

3.20.

pp. 118 - 119

500 1

XXX

2

1 2

$$1 : 2 = 3 : 1$$

K₁

0.98

K₂

0.90

0.01

(1)

1

(2) (1)

$$\begin{aligned}
 (1) \quad & \binom{1}{1} = \frac{1}{500} \cdot \frac{3}{4} = \frac{3}{2000} \quad \binom{1}{2} = \frac{1}{500} \cdot \frac{1}{4} = \frac{1}{2000} \quad \binom{0}{0} = \frac{499}{500} = \frac{1996}{2000} \\
 & \binom{0}{1} = \frac{3}{2000} \cdot \frac{98}{100} + \frac{1}{2000} \cdot \frac{90}{100} + \frac{1996}{2000} \cdot \frac{1}{100} = \frac{294 + 90 + 1996}{200000} = \frac{2380}{200000} \\
 & \binom{1}{1|} = \frac{\binom{1}{1 \cap}}{\binom{0}{1}} = \frac{294}{2380} \approx 0.1235 \\
 (2) \quad &
 \end{aligned}$$

3.21.

2023

pp. 122 - 124

$$0 < \rho < 1$$

$$\binom{0}{1} = \left\{ \begin{array}{l} 1 \quad (\rho = 1) \\ -1 \quad (\rho = -1) \end{array} \right.$$

=

$$(1) \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$(2) \quad \begin{bmatrix} 1 - \rho \\ 1 - \rho \end{bmatrix} \begin{bmatrix} 1 - \rho \\ 1 - \rho \end{bmatrix}$$

$$(3)$$

$$(4) \quad (3) \quad \Pr[X_1 + X_2 + X_3 \leq 2]$$

$$(1) \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - (1 - \rho) = 2 - 1 \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (2 - 1)^2$$

$$(2) \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}^2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{Cov}(X_1, X_2) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (2 - 1)[1 - (2 - 1)^2] = 4(1 - \rho)(2 - 1)$$

$$(3) \quad \text{Cov}(X_1, X_2) = 0 \quad \text{---} \quad 0 < \rho < 1 \quad = \frac{1}{2}$$

$$\binom{0}{1} = \binom{0}{1}, \binom{0}{1} = \binom{0}{1} \binom{0}{1} = \frac{1}{4}$$

$$(4) \quad \binom{0}{1}, \binom{0}{1} = (1, 1) \quad + \quad + \quad = 3 > 2$$

$$\Pr[X_1 + X_2 + X_3 \leq 2] = 1 - \Pr[X_1 + X_2 + X_3 > 2] = \frac{3}{4}$$

3.22.

2023

2024

pp. 125 - 126

2

(1)

(2) $\text{Cov}(,)$

(3)

	$= 1$	$= 10$	$= 100$
$= 1$			
$= 2$	0		
$= 3$			

(1) $8 = 1 = \frac{1}{8}$

(2)

- $[] = \frac{3}{8} 1 + \frac{2}{8} 2 + \frac{3}{8} 3 = 2$
- $[] = \frac{2}{8} 1 + \frac{3}{8} 10 + \frac{3}{8} 100 = \frac{83}{2}$
- $[] = \frac{664}{8} = 83$

$$\text{Cov}(,) = [] - [] [] = 83 - 2 \frac{83}{2} = 0$$

(3) $(= 2) = \frac{1}{4} \quad (= 1) = \frac{1}{4} \quad (= 2, = 1) = 0$
 $(= 2, = 1) \neq (= 2) (= 1)$

3.23.

2023

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1

$(+ 1)$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 6 \end{pmatrix} \begin{pmatrix} 1 \\ 6 \end{pmatrix}^{-1} = \frac{1}{6}$$

$2 \times 1 \quad 1 \times (-1)$

4.**4.1.**

2025

2 (2,)

$$\begin{aligned}
 (2,) \quad (=) &= \binom{2}{ } (1 -)^{2-} \\
 [] &= \sum_{=0}^2 (=) = (= 1) + 2 (= 2) \\
 &= \binom{2}{1} (1 -) + 2 \binom{2}{2} ^2 \\
 &= [2(1 -) + 2] = 2 \\
 \text{Var}() &= [(- [])^2] = [(- 2)^2] \\
 &= \sum_{=0}^2 (- 2)^2 (=) \\
 &= (0 - 2)^2 (= 0) + (1 - 2)^2 (= 1) + (2 - 2)^2 (= 2) \\
 &= 4 ^2 \binom{2}{0} (1 -)^2 + (1 - 2)^2 \binom{2}{1} (1 -) + 4(1 -)^2 \binom{2}{2} ^2 \\
 &= 8 ^2 (1 -)^2 + 2 (1 -)(1 - 2)^2 = 2 (1 -)
 \end{aligned}$$

4.2.

2024

p. 141

1 6

(1) 10 5

(2) 100 25

$$\begin{aligned}
 (1) \quad (10, \frac{1}{6}) \\
 (\geq 5) &= \sum_{=5}^{10} (=) = \sum_{=5}^{10} \binom{10}{ } \left(\frac{1}{6}\right) \left(\frac{5}{6}\right)^{10-} = \frac{1}{6^{10}} \sum_{'=0}^5 \binom{10}{'} 5 ^ \\
 &= \frac{1}{6^{10}} (1 ^1 + 10 ^5 + 45 ^25 + 120 ^3125 + 210 ^4625 + 252 ^53125) \\
 &= \frac{934926}{60466176} = 0.01546... \approx 0.0155
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad & (100, \frac{1}{6}) \quad [] = \frac{50}{3} \quad [] = \frac{5\sqrt{5}}{3} \\
 & (1 -) > = \frac{50}{3} > 5 \\
 & (100, (5\sqrt{5}/3)^2) \\
 & (\geq 25) \approx (> 24.5) = (> \frac{24.5 - 50/3}{5\sqrt{5}/3}) = (> \frac{23.5}{5\sqrt{5}}) \\
 & = (> 0.94\sqrt{5}) \approx (> 2.102) \approx 1 - 0.9821 = 0.0179
 \end{aligned}$$

4.3.

2023

2026

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$$2 \quad \text{Bin}(,) \quad = (-)$$

$$(1) \quad (| - | \leq 0.02) \geq 0.95$$

$$(2) \quad (1) \quad (2)$$

$$(3)$$

$$(3) \quad (1) \quad -$$

$$(4) \quad (2) \quad (3)$$

$$\begin{aligned}
 (1) \quad & | - | \leq 0.02 \quad 0.02 \\
 & (| - | \leq 0.02) \geq 0.95 \quad 0.02
 \end{aligned}$$

$$\begin{aligned}
 & 0.95 \\
 (2) \quad \text{Var}() &= \frac{(1 -)}{0.02^2} = \frac{0.02}{\sqrt{\text{Var}()}}
 \end{aligned}$$

$$(| - | \leq 0.02) = (| - | < \sqrt{\text{Var}()}) \geq 1 - \frac{1}{2} = 1 - \frac{(1 -)}{0.02^2} \geq 0.95$$

$$\begin{aligned}
 \frac{(1 -)}{0.02^2} &\leq 0.05 \quad \geq 50,000 (1 -) \\
 0^2 < (1 -) &\leq \left(\frac{1}{2}\right)^2 \quad \geq 12,500
 \end{aligned}$$

$$(3)$$

$$(, \text{Var}())$$

$$\begin{aligned}
 (| \bar{y} - \mu | \leq 0.02) &= \left(\left| \frac{\bar{y} - \mu}{\sqrt{\text{Var}(\bar{y})}} \right| \leq \frac{0.02}{\sqrt{\text{Var}(\bar{y})}} \right) = \left(|Z| \leq \frac{0.02}{\sqrt{\text{Var}(\bar{y})}} \right) \geq 0.95 \\
 &\quad (|Z| \leq 1.96) \approx 0.95 \\
 \frac{0.02}{\sqrt{\text{Var}(\bar{y})}} &\geq 1.96 \quad \geq \left(\frac{1.96}{0.02} \right)^2 (1 - \alpha) = 9,604 (1 - \alpha) \geq 2,401 \\
 (4)
 \end{aligned}$$

$$\begin{aligned}
 (| \bar{y} - \mu | \leq \delta) &\geq 1 - \\
 &= \sqrt{(1 - \alpha)}
 \end{aligned}$$

$$(| \bar{y} - \mu | \leq \delta) = (| \bar{y} - \mu | < -\delta) \geq 1 - (-\delta)^2$$

$$(| \bar{y} - \mu | \leq \delta) \geq 1 - (-\delta)^2 \geq 1 -$$

$$\geq (-\delta)^2 = \frac{(1 - \alpha)}{2} \geq \frac{(1 - \alpha)}{2} \geq \frac{1}{4} \cdot 2$$

$$(| \bar{y} - \mu | \leq \delta) = (| \bar{y} - \mu | \leq -\delta) = (|Z| \leq -\delta) = 2\Phi(-\delta) - 1$$

$$(| \bar{y} - \mu | \leq \delta) \geq 2\Phi(-\delta) - 1 \geq 1 -$$

$$\Phi\left(\sqrt{\frac{(1 - \alpha)}{2}}\right) \geq 1 - \frac{\alpha}{2}$$

$$\left(\frac{1 - \alpha}{2}\right) = \Phi\left(\sqrt{\frac{(1 - \alpha)}{2}}\right)$$

$$\sqrt{\frac{(1 - \alpha)}{2}} \geq \left(\frac{1 - \alpha}{2}\right) \geq \frac{(1 - \alpha)}{2} \left[\left(\frac{1 - \alpha}{2}\right)\right]^2 \geq \frac{1}{4} \left[\left(\frac{1 - \alpha}{2}\right)\right]^2$$

4.4.

2024

pp. 143 - 145, 153

(1) :

$$f(x) = \sqrt{\frac{1}{2}} e^{-\frac{(x-1)^2}{2}} \quad (-\infty < x < \infty)$$

$$f'(x) = -\frac{(x-1)}{2} \sqrt{\frac{1}{2}} e^{-\frac{(x-1)^2}{2}}$$

$$(2) \quad f''(x) = \frac{(x-1)^2 - 1}{2} \sqrt{\frac{1}{2}} e^{-\frac{(x-1)^2}{2}}$$

$$(3) \quad f'''(x) = \frac{(x-1)^3 - 3(x-1)}{2} \sqrt{\frac{1}{2}} e^{-\frac{(x-1)^2}{2}}$$

$$(1) \quad f'(x) = -\frac{(x-1)}{2} \sqrt{\frac{1}{2}} e^{-\frac{(x-1)^2}{2}} \quad f''(x) = \frac{(x-1)^2 - 1}{2} \sqrt{\frac{1}{2}} e^{-\frac{(x-1)^2}{2}}$$

$$(2) \quad f'''(x) = \frac{(x-1)^3 - 3(x-1)}{2} \sqrt{\frac{1}{2}} e^{-\frac{(x-1)^2}{2}}$$

$$(3) \quad f^{(4)}(x) = \frac{(x-1)^4 - 6(x-1)^2 + 3}{2} \sqrt{\frac{1}{2}} e^{-\frac{(x-1)^2}{2}}$$

$$0.5$$

4.5.

2022

pp. 121 - 122, 143

$$(1) \quad (170, 6^2)$$

$$(a) \quad \frac{1}{2} = \frac{1}{2}$$

$$(b) \quad \frac{1}{2} = \frac{1}{2}$$

$$(2) \quad \frac{1}{2} = \frac{1}{2}$$

$$f(x) = \frac{1}{2} \quad (0 \leq x \leq 2), \quad f(x) = \frac{1}{12} \quad (-3 \leq x \leq 9)$$

$$=$$

$$(1) \quad (a) \quad E[X] = E\left[\frac{1}{2} + \frac{1}{2}X\right] = \frac{1}{2}E[X] + \frac{1}{2}E[X] = \frac{1}{2} \cdot 170 + \frac{1}{2} \cdot 170 = 170$$

$$(b) \quad \text{Var}(X) = \text{Var}\left(\frac{1}{2} + \frac{1}{2}X\right) = \left(\frac{1}{2}\right)^2 \text{Var}(X) + \left(\frac{1}{2}\right)^2 \text{Var}(X) = \frac{1}{4} \cdot 6^2 + \frac{1}{4} \cdot 6^2 = 18$$

$$(2) \quad E[X^2] = \int_0^2 x^2 f(x) dx = \frac{1}{2} \int_0^2 x^2 dx = \frac{1}{4} [x^3]_0^2 = \frac{(2)^3}{2} = 4$$

$$E[X^2] = \int_{-3}^9 x^2 f(x) dx = \frac{1}{12} \int_{-3}^9 x^2 dx = \frac{1}{24} [x^3]_{-3}^9 = \frac{1}{24} (9^3 - (-3)^3) = 3$$

$$[] = [] = 3 = 3$$

4.6.

2024

pp. 147 - 151

$$(1) \quad (, 2)$$

$$(a) \quad (- \leq \leq +)$$

$$(b) \quad (- 2 \leq \leq + 2)$$

$$(c) \quad (- 3 \leq \leq + 3)$$

$$(2) \quad ($$

$$(3)$$

$$(a) \quad (0.01)$$

$$(b) \quad \Phi(0) = 0.01 \quad 0$$

$$(c) \quad (0.005)$$

$$(1) \quad (+ \leq \leq +) = \Phi() - \Phi() \quad \Phi() - \Phi(-) = 2\Phi() - 1$$

$$(a) \quad (- \leq \leq +) = \Phi(1) - \Phi(-1) = 0.6826$$

$$(b) \quad (- 2 \leq \leq + 2) = \Phi(2) - \Phi(-2) = 0.9544$$

$$(c) \quad (- 3 \leq \leq + 3) = \Phi(3) - \Phi(-3) = 0.9974$$

$$(2) \quad 3 \quad [- 3 , + 3] \quad 3$$

$$(3) \quad (a) \quad 2.327$$

$$(b) \quad -2.327$$

$$(c) \quad 2.575$$

4.7.

2024

pp. 149 - 151

$$(1) \quad (0, 1)$$

$$(| | \geq) = 2 (\geq)$$

$$(2) \quad (, 2)$$

$$(|\bar{X} - \mu| \geq \frac{1}{4})$$
$$4$$
$$(3) \quad \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} dt = 1$$
$$(|\bar{X} - \mu| \geq \frac{1}{4}) \leq 0.02$$

$$(1) \quad f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad f(-x) = f(x)$$
$$(|X| \geq a) = \int_{-\infty}^{-a} f(x) dx + \int_a^{\infty} f(x) dx = \int_a^{\infty} f(x) dx + \int_a^{\infty} f(x) dx = 2 \int_a^{\infty} f(x) dx$$
$$(2) \quad (|\bar{X} - \mu| \geq \frac{1}{4}) = (|X| \geq \frac{1}{4}) \approx 0.8026 \approx 0.803$$
$$(3) \quad [X] = \lfloor X \rfloor = \sqrt{X} =$$
$$(|\bar{X} - \mu| \geq \frac{1}{4}) = (|X| \geq \frac{\sqrt{X}}{4}) \leq 0.02$$
$$(X \leq 2.33) = 0.9901 \quad (|X| > 2.33) \approx 0.0198 \approx 0.02$$
$$\frac{\sqrt{X}}{4} \geq 2.33 \quad \Rightarrow (2.33 \cdot 4)^2 = 86.56$$
$$= 87$$

4.8.

2024

2026

pp. 149 - 150

$$4$$
$$350,198 \quad 66.31$$
$$23.55 \quad (66.31, (23.55)^2)$$
$$(1) \quad (X \geq 80)$$
$$(2) \quad (3-a)$$
$$(3) \quad 80 \quad 95$$
$$(4) \quad 20\%$$

$$\begin{aligned}
 (1) \quad & \mu = 350,198 \quad \sigma = 66.31 \quad \sigma^2 = 23.55 \quad \left(\frac{1}{\sigma^2} \right) \\
 & = \frac{1}{23.55} \quad (0,1) \quad \left(\frac{1}{\sigma^2} \geq 80 \right) = \left(\frac{1}{\sigma^2} \geq \right. \\
 & \left. \frac{80 - \mu}{\sigma} \right) \approx \left(\frac{1}{\sigma^2} \geq 0.58 \right) \quad \left(\frac{1}{\sigma^2} = 80 \right) \approx 0 \quad \left(\frac{1}{\sigma^2} \geq \right. \\
 & 80) = \left(\frac{1}{\sigma^2} \geq 0.58 \right) = 1 - \Phi(0.58) \quad \Phi(0.58) \approx 0.7910 \\
 & \left(\frac{1}{\sigma^2} \geq 80 \right) \approx 1 - 0.7910 = 0.2090
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad & \frac{1}{\sigma^2} \quad 80
 \end{aligned}$$

$$\begin{aligned}
 (3) \quad & [80, 90] \quad (80 \leq \frac{1}{\sigma^2} \leq 90) \\
 & (80 \leq \frac{1}{\sigma^2} \leq 90) = \left(\frac{1}{\sigma^2} \leq \frac{90 - \mu}{\sigma} \right) - \left(\frac{1}{\sigma^2} \leq \frac{80 - \mu}{\sigma} \right) \\
 & = \left(\frac{1}{\sigma^2} \leq \frac{90 - 66.31}{2} \right)
 \end{aligned}$$

$$\begin{aligned}
 & -\frac{-120}{10} \approx 1.65 & \approx 120 - 1.65 \times 10 = 103.5 \\
 (2) \quad (100, 15^2) & \quad (\geq) = 1 - \Phi\left(\frac{103.5 - 100}{15}\right) \approx 1 - 0.5910 = 0.4090
 \end{aligned}$$

4.10.

$$(42 \leq x \leq 78) = (|x - 60| < 18) = (|x - 60| < 3 \times 6) \geq 1 - \frac{1}{3^2} = \frac{8}{9}$$

(2)

$$=$$

$$= (42 \leq x \leq 78) \geq \frac{8}{9} \times 178 = 177.7...$$

(3)

$$(x, x^2)$$

$$(42 \leq x \leq 78) = (-3 \leq x \leq 3) = 0.9974$$

5.1.

pp. 168 - 169

2

$$\begin{aligned} (x, y) &= (x, y) = (x, y) (1 - \epsilon)^{-1} (1 - \epsilon) \\ &= (x, y) (1 - \epsilon)^{-1} (1 - \epsilon) \\ &= (x, y) (1 - \epsilon)^{-1} (1 - \epsilon) \rightarrow \end{aligned}$$

pp. 93, 168 - 169

1	ATM	ATM	1	40
	ATM		1	60
		ATM		
		/		
60				
75				
90				
105				
120				

$$\frac{1}{1 - \frac{1/90}{1 - 60/90}} h = \frac{1}{30} h = 120 \text{ s}$$

pp. 166 - 169

$$(1) \quad 3 \quad 5$$

$$(2) \quad 1 \quad 2 \quad 1 \quad 4$$

$$(1) \quad 5 \quad (5, 0.3)$$

$$(X \geq 1) = 1 - (X = 0) = 1 - (0.7)^5 = 1 - 0.16807 = 0.83193$$

$$(2) \quad 1 \quad (2)$$

$$(X \geq 4) = 1 - (0 \leq X \leq 3) = 1 - \sum_{k=0}^3 \binom{5}{k} (0.3)^k (0.7)^{5-k} = 1 - \frac{19}{3} (0.3)^2$$

$$\approx 1 - \frac{19}{3} \cdot \frac{18}{133} = \frac{57}{399} \approx 0.1429$$

$$X^2 = 1 + \frac{2}{(1-1) + \frac{1}{3 + \frac{1}{5 + \frac{1}{7 + \frac{1}{9 + \dots}}}}} \approx 1 + \frac{2}{0 + \frac{1}{3 + \frac{1}{5 + \frac{1}{7 + 0}}}} = \frac{18}{399} \quad (5.1)$$

5.4.

2025

4

$$(1) \quad 1 \quad 3$$

2

$$(2) \quad (X = 3) = 5 \quad (X = 5)$$

$$(X \leq 3)$$

$$(1) \quad (3)$$

$$(X \geq 2) = 1 - (0 \leq X \leq 1) = 1 - \sum_{k=0}^1 \binom{5}{k} (0.3)^k (0.7)^{5-k} = 1 - \frac{4}{3}$$

$$\approx 1 - 4 \cdot \frac{221}{4439} = \frac{3555}{4439} \approx 0.8009$$

$$^3 \approx 1 + \frac{6}{(2-3) + \frac{9}{6 + \frac{9}{10 + \frac{9}{14 + \frac{9}{18 + 0}}}}} = \frac{4439}{221} \tag{5.2}$$

$$(2) \quad (\quad = 3) = \quad - \frac{3}{3!} = \quad - \frac{3}{6} \qquad 5 \quad (\quad = 5) = 5 \quad - \frac{5}{5!} = \quad - \frac{3}{6} \left(\frac{2}{2} \right)^2 \qquad = 2$$
$$(2)$$

$$(\quad \leq 3) = \quad^{-2} (1 + 2 + 2 + \frac{4}{3}) \approx \frac{18}{133} \quad \frac{19}{3} = \frac{342}{399} \approx 0.8751$$
$$(5.1)$$

5.5.

2023

pp. 166 - 169

1%200

(1)

(2)

(3)3

(2)(3)

$$(1) \quad (\quad = \quad) = (\quad) \left(\frac{1}{100} \right) \left(\frac{99}{100} \right)^{200 -} \approx \quad^{-2} \frac{2}{!}$$
$$(2) \quad (\quad = 0) \approx \quad^{-2} \approx \frac{18}{133} \approx 0.1353$$
$$(3) \quad (\quad \geq 3) = 1 - \quad (0 \leq \quad \leq 2) = 1 - \quad^{-2} (1 + 2 + 2) \approx 1 - 5 \quad \frac{18}{133} = \frac{43}{133} \approx 0.3233$$

5.6.

2024

pp. 153 - 154, 166 - 169

4

(1)24063845

(2)0.1%5003

$$\begin{aligned}
 (1) \quad & = 240 \quad = \frac{1}{6} \quad 6 \quad (,) \quad = \\
 & 40 \quad (1 -) = 200 \quad 5 \\
 & (, (1 -)) = (40, \frac{100}{3}) \\
 & (38 \leq \leq 45) \approx (37.5 \leq \leq 45.5) = (\frac{37.5 - 40}{\frac{10}{\sqrt{3}}} \leq \leq \frac{45.5 - 40}{\frac{10}{\sqrt{3}}}) \\
 & = (-0.25 \sqrt{3} \leq \leq 0.55 \sqrt{3}) \approx (-0.43 \leq \leq 0.95) \\
 & = 0.8289 - 0.3336 = 0.4953 \\
 (2) \quad & = 500 \quad = \frac{1}{1000} \quad (,) \\
 & \quad \quad \quad () \\
 & = \quad = \frac{1}{2} \\
 & (\geq 3) = 1 - (0 \leq \leq 2) = 1 - {}^{-\frac{1}{2}} (1 + \frac{1}{2} + \frac{1}{8}) = 1 - {}^{-\frac{1}{2}} \frac{13}{8} \\
 & \approx 1 - \frac{12}{12} \frac{2^2 - 6}{2^2 + 6} \frac{2 + 1}{2 + 1} \frac{13}{8} = 1 - \frac{37}{61} \frac{13}{8} = \frac{7}{488} \approx 0.0143 \\
 & \sqrt{-} \approx 1 + \frac{2}{(2 - 1) + \frac{1}{6 + 0}} = \frac{12}{12} \frac{2 + 6}{2 - 6} \frac{+ 1}{+ 1} \quad (5.3)
 \end{aligned}$$

5.7.

2026

$$= 0.001$$

$$= 500$$

$$^{-0.5} \approx 0.6065, \quad \sqrt{\frac{1}{0.5}} \approx 0.7071$$

$$(1) \quad (a)$$

$$(b) \quad [] \quad \text{Var}()$$

$$(2)$$

$$(a)$$

$$(b) \quad (\geq 3)$$

(3)

(a)

(b) (≥ 3)

(4)

(a) (2) (3)

(b)

(1) (a) $(500, 0.001)$

(b) $[] = 0.5 \quad \text{Var}() = (1 -) = 0.4995$

(2) (a)

$$(5) \quad (4)$$

$$(6) \quad = \quad \rightarrow \infty \quad (4) \quad (\& ,)$$

$$(;) \quad \lim_{\rightarrow \infty} (1 + \text{---}) =$$

$$(1) \quad (1 -) \quad (1 -) = {}^3(1 -)^2$$

$$(2)$$

$$(3) \quad [] = \sum_{=0}^{-1} (1 -) + = \frac{\text{---}}{1 - } (1 -)$$

$$(4) \quad (\& ,) = () \quad (1 -)^{-}$$

$$(5) \quad [] = \sum_{=0} \quad (\& ,) =$$

$$(6) \quad =$$

$$\lim_{\rightarrow \infty} () \quad (1 -)^{-}$$

$$= \lim_{\rightarrow \infty} \frac{\text{---}!}{!(\text{---})!} \quad (1 -)^{-}$$

$$= \lim_{\rightarrow \infty} \frac{1}{\text{---}!} \quad (\text{---} - 1)(\text{---} - 2) (\text{---} - + 1)(\text{---}) (1 - \text{---})^{-}$$

$$= \lim_{\rightarrow \infty} \frac{\text{---}}{\text{---}!} (1 - \frac{1}{\text{---}}) (1 - \frac{2}{\text{---}}) \quad (1 - \frac{\text{---} - 1}{\text{---}}) (1 - \text{---})^{-} (1 + \frac{\text{---}}{\text{---}})$$

$$= \frac{\text{---}}{\text{---}!}^{-}$$

6.

6.1.

2022

p. 174

2

(1)

(2)

(1)

(2)

6.2.

2025

2

$1, 2, \dots$

(1)

$1, 2, \dots$

(2)

(3)

$$= 1 + 2 + \dots$$

(4)

(1)

(2) $> 0 : \lim_{\rightarrow \infty} (| - | <) = 1$

(3) $= \frac{2}{2} = 1$

$$= \frac{2}{2} = 1$$

(4)

6.3.

2025

p. 179

$$\begin{array}{lcl}
 & 1180 & 20 \\
 (1) \ 25 & & 1170 \\
 (2) & & 0.9 \quad 1175
 \end{array}$$

$$\begin{array}{l}
 (1) \ 25 \quad \text{---} \quad \begin{array}{c} 1, \dots, 25 \\ (1180, \frac{20^2}{25}) \end{array} \quad 1180 \quad 20^2 \\
 \\
 (\bar{} > 1170) = \left(> \frac{1170 - 1180}{4} \right) = \left(> -2.5 \right) = 0.9938 \\
 \\
 (2) \quad \text{---} \quad \left(1180, \frac{20^2}{25} \right) \\
 \\
 (\bar{} > 1175) = \left(> \frac{1175 - 1180}{\frac{\sqrt{}}{20}} \right) = \left(> -\frac{\sqrt{}}{4} \right) \\
 \\
 \left(> -1.29 \right) \approx 0.9015 > 0.9 \\
 \\
 -1.29 \geq -\frac{\sqrt{}}{4} \quad \sqrt{} \geq 1.29 \times 4 = 5.16 \quad \geq (5.16)^2 = 26.6256 \\
 \geq 27
 \end{array}$$

6.4.

2025

$$\begin{array}{lcl}
 & 1.2 \text{ kg} & 0.1 \text{ kg} \\
 100 & & 122.5 \text{ kg}
 \end{array}$$

$$\begin{array}{l}
 \begin{array}{c} 1, \dots, 100 \\ [] = 100 \times 1.2 \text{ kg} = 120 \text{ kg} \\ (120 \text{ kg}, 1 \text{ kg}) \end{array} \quad \begin{array}{c} = 1 + \dots + 100 \\ [] = \sqrt{100} \times 0.1 \text{ kg} = 1 \text{ kg} \end{array} \\
 \\
 (< 122.5 \text{ kg}) = \left(< \frac{122.5 - 120 \text{ kg}}{1 \text{ kg}} \right) = \left(< 2.5 \right) = 0.9938
 \end{array}$$

6.5.

2022

pp. 202 - 205

$$\begin{aligned}
 (1) \quad & \begin{matrix} & 2 & & & 3 & & & 1 & 2 & 3 \\ & & & & & & & & & \end{matrix} \\
 & \cdot \\
 (a) \quad & x_1 = x_1 + x_2 - x_3 \\
 (b) \quad & x_2 = \frac{x_1 + 2x_2 + x_3}{4} \\
 (c) \quad & x_3 = \frac{x_1 + x_2 + x_3}{3}
 \end{aligned}$$

$$(2) \quad \{x_1, x_2, x_3\}$$

$$\begin{aligned}
 (1) \quad & \\
 (a) \quad & (x_1) = (x_1) + (x_2) - (x_3) = + - = \\
 (b) \quad & (x_2) = \frac{1}{4} (x_1) + \frac{2}{4} (x_2) + \frac{1}{4} (x_3) = \frac{1}{4} + \frac{2}{4} + \frac{1}{4} = \\
 (c) \quad & (x_3) = \frac{1}{3} (x_1) + \frac{1}{3} (x_2) + \frac{1}{3} (x_3) = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \\
 (2) \quad & \\
 (a) \quad & (x_1) = 1^2 x_1^2 + 1^2 x_2^2 + (-1)^2 x_3^2 = 3 x^2 \\
 (b) \quad & (x_2) = \left(\frac{1}{4}\right)^2 x_1^2 + \left(\frac{2}{4}\right)^2 x_2^2 + \left(\frac{1}{4}\right)^2 x_3^2 = \frac{3}{8} x^2 \\
 (c) \quad & (x_3) = \left(\frac{1}{3}\right)^2 x_1^2 + \left(\frac{1}{3}\right)^2 x_2^2 + \left(\frac{1}{3}\right)^2 x_3^2 = \frac{1}{3} x^2
 \end{aligned}$$

6.6.

2023

2023

2025

pp. 176 - 180, 202 - 205

$$\begin{aligned}
 (1) \quad & \begin{matrix} 0 & 1.2 & 3 \end{matrix} \\
 & \{-1, 0, 1\} \\
 (a) \quad & \\
 (b) \quad & \\
 (c) \quad & \\
 (2) \quad &
 \end{aligned}$$

- (3)
- (4) $\Theta = (\theta_1, \theta_2, ..., \theta_3)$
- (a)
- (b)
- (c)

- (5)
- 3

- (1) (a) 1.2
- (b) $\frac{1}{3}[(-1)^2 + 0^2 + 1^2] = \frac{2}{3}$
- (c) $\frac{3}{2} \times \frac{2}{3} = 1$
- (2)
- (3)
- (4) (a) $\hat{\theta}$
- $[\hat{\theta}] =$
- (b)
- $> 0 : \lim_{\rightarrow \infty} (| \hat{\theta} - \theta | < \epsilon) = 1$
- (c) $\hat{\theta}$
- $\hat{\theta}' : \hat{\theta} \leq \hat{\theta}'$
- (5)
- $\rightarrow \infty$

6.7.

1000

10

32, 86, 78, 49, 48, 77, 97, 49, 51, 74

1000

$$\bar{x} = \frac{1}{10}(32 + 86 + 78 + 49 + 48 + 77 + 97 + 49 + 51 + 74) = 64.1$$
$$\overline{x^2} = \frac{1}{10}(32^2 + 86^2 + 78^2 + 49^2 + 48^2 + 77^2 + 97^2 + 49^2 + 51^2 + 74^2) = 4502.5$$
$$s^2 = \overline{x^2} - (\bar{x})^2 = 4502.5 - (64.1)^2 = 4502.5 - 4108.81 = 393.69$$
$$s^2 = \frac{10}{9} s^2 = \frac{10}{9} \times 393.69 \approx 437.43$$

6.8.

2025

pp. 179, 216

- (1) .
- (2)
- (3) 0 95%

- (4) = 0.05 = 1900 (76 ≤ ≤
- 114)

(1) (,)

(2)

$$\left(\frac{1}{2}, \left(1 - \frac{1}{2}\right) \right)$$

$$(3) \quad 2 \left(\frac{1}{2}\right) \sqrt{\frac{0(1-0)}{10000}}$$

$$\left(\frac{1}{2}\right) \sqrt{\frac{0(1-0)}{10000}} = \frac{1}{2} \left(\frac{1}{2}\right) \sqrt{\frac{0(1-0)}{10000}}$$

$$= 4$$

$$(4) \quad = 1900 \cdot 0.05 = 95 \quad (1 - 0.05) = 1900 \cdot 0.05 \cdot 0.95 = 90.25 = 9.5^2$$

$$(95, 9.5^2)$$

$$(76 \leq \leq 114) \approx (75.5 < < 114.5) = \left(\frac{75.5 - 95}{9.5} < < \frac{114.5 - 95}{9.5} \right)$$

$$\approx (-2.05 < < 2.05) \approx 0.9596$$

6.9.

2022

pp. 180 - 182

10,000

$$3,500 = \frac{3500}{10000} = 0.35$$

(1)

$$(2) \quad 2$$

(1)

$$\left(\frac{1}{2}, \left(1 - \frac{1}{2}\right) \right)$$

95%

$$\left[0.35 - 1.96 \sqrt{\frac{0.35 \cdot 0.65}{10000}}, 0.35 + 1.96 \sqrt{\frac{0.35 \cdot 0.65}{10000}} \right] \approx [0.3407, 0.3593]$$

0.35

$$(2) \quad 10,000 \times 2^2 = 40,000$$

6.10.

2024

2024

2025

pp. 224 - 225

$$(1) \quad (1,)$$

1, 2, ...,

(a)

(b)

(2)

A

(a)

 $= -$

A

 $100(1 -)\%$

(b)

(3)

S

TV

T

140

32

T

(a)

95%

(b)

3%

0.95

(c)

$$(1) \quad (a) \quad [] = , \quad \text{Var}() = \frac{(1 -)}{2}$$

$$(b) \quad = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} \quad (, (1 -))$$

$$[] = , \quad \text{Var}() = (1 -)$$

$$[] = [-] = \frac{1}{2} \quad [] = - = ,$$

$$\text{Var}() = \text{Var}(-) = \frac{1}{2} \text{Var}() = \frac{(1 -)}{2} = \frac{(1 -)}{2}$$

$$(2) \quad (a) \quad [- (\frac{1}{2}) \sqrt{\frac{(1 -)}{2}}, + (\frac{1}{2}) \sqrt{\frac{(1 -)}{2}}]$$

(b)

$$(3) \quad (a) \quad [\frac{32}{140} \pm 1.96 \sqrt{\frac{32 \times (140 - 32)}{140^3}}] \approx [0.2286 \pm 0.06956] \approx [0.1590, 0.2982]$$

$$(b) \quad \approx 0.2286$$

$$| - | = 1.96 \sqrt{\frac{0.2286 \times 0.7714}{2}} \leq 0.03$$

753

(c)

$$(1 - \frac{1}{4}) \leq \frac{1 + (1 - \frac{1}{4})^2}{2} = \frac{1}{4}$$

$$AM \geq GM$$

$$\geq \left(\frac{1.96}{0.03}\right)^2 \times \frac{1}{4} \approx 1067.1$$

1068

6.11.

2024

pp. 223 - 225

(1) $\hat{\alpha} = 0.8$ $\hat{\beta} = 0.9$ $\hat{\gamma} = 0.99$ $\hat{\delta} = 0.02$

(2) $\hat{\alpha} = 0.03$ $\hat{\beta} = 0.95$ $\hat{\gamma} = 0.99$ $\hat{\delta} = 0.02$

$$(1)$$

$$= \frac{\hat{\beta}}{\sqrt{(1-\beta)/n}}$$

$$(0, 1) \quad (|t| < 2.576) \approx 0.99$$

$$(|r - \rho| < 2.576 \sqrt{\frac{(1 - \rho^2)}{n}}) = 0.99$$

$$|\hat{\mu} - \mu| < 0.02$$
$$2.576 \sqrt{\frac{(1 - \alpha)}{n}} \leq 0.02 \qquad \geq \left(\frac{2.576}{0.02}\right)^2 (1 - \alpha)$$
$$[0.8, 0.9]$$
$$(1 - \alpha) \leq 0.8 \times 0.2 = 0.16$$
$$\geq \left(\frac{2.576}{0.02}\right)^2 \times 0.16 \approx 2654.3$$
$$2655$$

(2)

$$(1 - \alpha) \leq \frac{1}{4}$$
$$\geq \left(\frac{1.96}{0.03}\right)^2 \times \frac{1}{4} \approx 1067.1$$
$$1068$$

6.12.

2025

2

1 -

(;)

(1) (;) () .

(2) (;) .

(3) () .

(4) .

(5) .

(a) (4) ?

(b) (4) ?

(c) ~~(4)~~ $\xrightarrow{\quad\quad\quad} \frac{?}{?} \leftarrow$

$$(1) \quad \binom{n}{k} = \binom{n}{n-k} (1-x)^{-n} x^k$$

$$\begin{aligned}
 (2) \quad [x]^n &= \sum_{k=0}^n \binom{n}{k} x^k \\
 &= \sum_{k=0}^n \binom{n}{k} (1-x)^{-n} x^k \\
 &= \sum_{k=0}^n \frac{n!}{k!(n-k)!} (1-x)^{-n} x^k \\
 &= \sum_{k=0}^n \frac{(n-k)!}{(n-k)! \{(n-k) - (n-k)\}!} x^{-1} (1-x)^{(n-k)-(n-k)} \\
 &= \sum_{k'=0}^{n-1} \frac{(n-1-k')!}{k'! \{(n-1-k') - k'\}!} x^{-1} (1-x)^{(n-1-k')-k'} \\
 &= \sum_{k'=0}^{n-1} \binom{n-1-k'}{k'} x^{-1} (1-x)^{(n-1-k')-k'} \\
 &=
 \end{aligned}$$

$$\begin{aligned}
 (3) \quad \prod_{k=1}^n & \\
 \binom{n}{k} = \log \binom{n}{k}; &= \log \binom{n}{k} (1-x)^{-n} = \log \binom{n}{k} + \log (1-x)^{-n} + (n-k) \log(1-x)
 \end{aligned}$$

$$(4) \quad \frac{d}{dx} \binom{n}{k} = -\frac{n-k}{1-x} = 0$$

$$\hat{=} = _$$

$$(5) \quad (a) \quad \lceil _ \rceil = \lfloor _ \rfloor = \frac{1}{_} \quad \lceil _ \rceil = \frac{1}{_} (_) =$$

$$(b) \quad \lim_{\rightarrow \infty} \hat{=} = \lim_{\rightarrow \infty} _ =$$

$$(c) \quad \mathbf{AI}$$

6.13.

2025

$$(\mathbf{y}, \mathbf{y}^2)$$

(1)

(2)

$$(1) \quad \begin{aligned} &(\mathbf{y}, \mathbf{y}^2) \\ &[\mathbf{y}] = \mathbf{y}_1 \quad \hat{\mathbf{y}}_1 = \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i = \bar{\mathbf{y}} \\ &\quad \quad \quad \hat{\mathbf{y}} = \bar{\mathbf{y}} \end{aligned}$$

$$\begin{aligned} &\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n \\ &\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n \end{aligned} \quad (\mathbf{y}, \mathbf{y}^2)$$

$$(\mathbf{y}; \mathbf{y}) = \sqrt{\frac{1}{2}} \exp\left[-\frac{(\mathbf{y} - \bar{\mathbf{y}})^2}{2}\right]$$

$$\begin{aligned} -\log(\mathbf{y}) &= \sum_{i=1}^n -\log(\mathbf{y}; \mathbf{y}) \\ &= \sum_{i=1}^n -\log\left[\sqrt{\frac{1}{2}} \exp\left[-\frac{(\mathbf{y}_i - \bar{\mathbf{y}})^2}{2}\right]\right] \\ &= \sum_{i=1}^n \left[\log \sqrt{\frac{1}{2}} - \frac{(\mathbf{y}_i - \bar{\mathbf{y}})^2}{2}\right] \\ &= \sum_{i=1}^n \frac{-}{2} \\ &= \frac{-}{2} = 0 \end{aligned}$$

$$= - \quad \hat{\mathbf{y}} = \bar{\mathbf{y}}$$

$$(2) \quad \begin{aligned} &\mathbf{y}_1 \quad \mathbf{y}_2 \quad \mathbf{y}_1 \quad \mathbf{y}_2 \\ &\mathbf{y}_1 = [\mathbf{y}], \quad \mathbf{y}_2 = [\mathbf{y}^2] \end{aligned}$$

$$\mathbf{y}_2 = [\mathbf{y}^2] - [\mathbf{y}]^2 = \mathbf{y}_2 - \mathbf{y}_1^2$$

$$\begin{aligned}
 \hat{\mu}_1 &= \frac{1}{n} \sum_{i=1}^n x_i = \bar{x} \\
 \hat{\mu}_2 &= \hat{\mu}_1^2 = \bar{x}^2 \\
 \hat{\sigma}^2 &= \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2 \\
 &= \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2 \\
 \log L(\hat{\mu}, \hat{\sigma}) &= \sum_{i=1}^n \log \left(\frac{1}{\sigma \sqrt{2\pi}} \exp \left[-\frac{(x_i - \mu)^2}{2\sigma^2} \right] \right) \\
 &= \sum_{i=1}^n \left[-\log \left(\frac{1}{\sigma \sqrt{2\pi}} \right) - \frac{(x_i - \mu)^2}{2\sigma^2} \right] \\
 &= \sum_{i=1}^n \left[-\frac{1}{2} \log(2\pi) - \frac{(x_i - \mu)^2}{2\sigma^2} \right] \\
 &= \sum_{i=1}^n \left[-\frac{1}{2} \log(2\pi) - \frac{(x_i - \mu)^2}{2\sigma^2} \right] \\
 &= -\frac{n}{2} \log(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \\
 \hat{\mu} &= \bar{x} \\
 \hat{\sigma}^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = s^2
 \end{aligned}$$

6.14.

2025

(1)

(2)

/

2

(3)

/

(1)

(2)

$$\text{Var}(\overline{X}) = \frac{\sigma^2}{n}$$

$$\frac{\sigma^2}{n}$$

(a)

$$= \frac{\sigma^2}{n}$$

$$\leq 5\%$$

$$10000$$

$$5$$

7.

- ① 仮説の検定
- ② 統計量の検定
- ③ 優位水準と棄却域の設定
- ④ 帰無仮説の棄却・選択

7.1.

2022

2023

pp. 227 - 231

(1)

(2)

(3)

(4)

5%

(3)

5%

(5)

(1)

(1)

(6)

(6)

(7)

• (8:)

• (9:)

7.2.

2022

pp. 227 - 234

(1)

(2)

7.3.

2023						pp. 2235 - 236
			8000			
		25			7700	
				5%		
			640			

7.4.

2023						pp. 238 - 240
	T					
S		365			6	
			330, 350, 385, 321, 340, 365			
				5%		

7.5.

2025						pp. 238 - 240
	T					
S		360			6	
			350, 355, 385, 390, 370, 365			

- (1)10%
- (2)5%

7.6.

2023	2024	2024	2025			pp. 235 - 236
(1)			(, 49)	49		
	$_0 : = 4,$	$_1 : > 4$				
	(a)	0.05	= 6	2		

(b)

0.01

= 7

2

(2)

2

0.2

0.1

$\frac{1}{2} : \frac{1}{5} = 5$

0.05

7.7.

2022

pp. 241 - 242

(1)

2

A

B

2

10

12

A

2.04

0.03

B

2.02

0.02

2

5%

2

(2)

5

1

5%

7.11.

2023

pp. 252 - 253

- 10
- $0 : = \frac{1}{2}$
- 0
- (1)
- (2)
- (a)
- $1 : < \frac{1}{2}$
- 5%
- (b)
- $1 : \neq \frac{1}{2}$
- 2%

7.12.

2023

pp. 252 - 253

- 5
- 1
- (1)
- 5
- 5
- 0
- (a)
- $= \frac{1}{2}$
- (b)
- $= \frac{2}{3}$
- (2)
- $\geq \frac{1}{2}$
- (a)
- 3
- $= \frac{1}{2}$
- (b)
- $= \frac{3}{4}$
- 3

7.13. 2

256

116

5%

- (1)
- 2

(2)

2

10%

7.14.

2023

2025

pp. 252 - 253

0

:

3

3

0

1

2

0

<

3

3

1

0

(1)

1

(2)

2

(3)

2

1

3

7.15.

2024

pp. 252 - 253

3

A

B

C

A

3

B

2

C

1

1

(1)

A

A

1

16

B

B

1

9

C

C

1

25

(2)

960

640

(1)

5%

52 of 52